Lecture 22: Grover's Search Algo

- Grover's Search Algo:
  ↳ Finds entry in an unordered database of size N is $\sim\sqrt{N}$ queries using quantum interference.

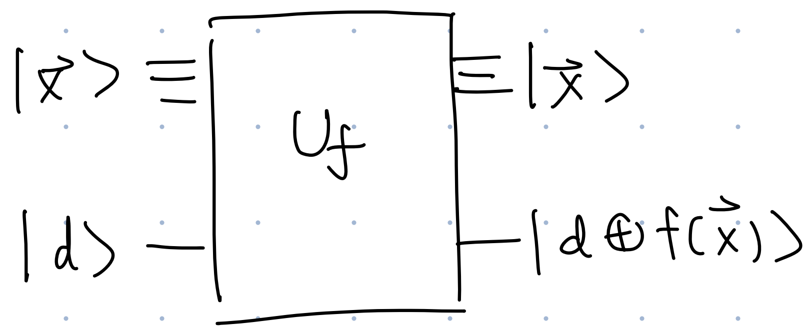
$\vec{x}$		
000	$\vec{y}(000)$	$\vec{z}(000)$
001	$\vec{y}(001)$	⋮
010	⋮	⋮
011	⋮	⋮
100	⋮	⋮
101	⋮	⋮
111	$\vec{y}(111)$	$\vec{z}(111)$
ordinal #	name	phone number.

$f : \{0,1\}^n \to \{0,1\}$.

$f(\vec{x}) = 1$ if $\vec{y}(\vec{x}) = \vec{c}$ ← name I am searching
$= 0$ otherwise.

↳ have people made these oracles?

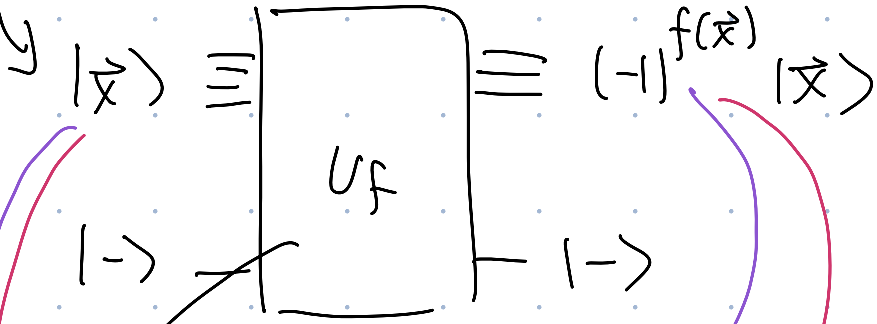
How does that work?



$$|\psi_{in}\rangle = |d\rangle \otimes |\vec{x}\rangle$$

$$|\psi_{out}\rangle = |d \oplus f(\vec{x})\rangle \otimes |\vec{x}\rangle$$

if $|d\rangle = |-\rangle \Rightarrow |\psi_{out}\rangle = |-\rangle \otimes (-1)^{f(\vec{x})} |\vec{x}\rangle$



if in computational basis → global phase.

placed in superposition → relative phase.

An $(n+1)\times(n-1)$ unitary matrix.

↳ Since it does nothing to $|-\rangle$, we can focus on the upper $n\times n$ part.

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\underline{U_f = I_{n\times n} - 2|\vec{y}\rangle\langle\vec{y}|}$$ → this is also unitary

$$U_f|y\rangle = -|y\rangle$$

$$U_f|x\rangle = |x\rangle.$$

$$|x\rangle \neq |y\rangle$$

Now put the upper register in superposition:

$$|\Phi_0\rangle = H^{\otimes n}|\vec{0}\rangle = (H|0\rangle)^{\otimes n} = \sum_{j=0}^{2^n-1} \frac{1}{\sqrt{2^n}}|b_j\rangle$$

$$= \frac{1}{\sqrt{2^n}} \sum_{\vec{x}} |\vec{x}\rangle$$

$$U_f|s\rangle = \frac{1}{\sqrt{2^n}} \sum_{\vec{x}\neq\vec{y}} |\vec{x}\rangle - \frac{1}{\sqrt{2^n}}|\vec{y}\rangle$$

$$=$$

Define:

$$U_g = I - 2|\vec{0}\rangle\langle\vec{0}|$$

Define $G \equiv -H^{\otimes n} U_g H^{\otimes n}$

$$= -H^{\otimes n}(I - 2|\vec{0}\rangle\langle\vec{0}|)H^{\otimes n}$$

$$= -H^{\otimes n} I H^{\dagger\otimes n} + 2H^{\otimes n}|\vec{0}\rangle\langle\vec{0}|H^{\dagger\otimes n}$$

$$= -I + 2|\Phi_0\rangle\langle\Phi_0|$$

$$= -(I - 2|\Phi_0\rangle\langle\Phi_0|)$$

Grover: apply $R = G U_f$ repeatedly

$$R = -\left(I - 2|\Phi_0\rangle\langle\Phi_0|\right)\left(I - 2|y\rangle\langle y|\right)$$

$$R|y\rangle = +\left(I - 2|\Phi_0\rangle\langle\Phi_0|\right)|y\rangle$$

$$R|\Phi_0\rangle = -\left(I - 2|\Phi_0\rangle\langle\Phi_0|\right)|\Phi_0\rangle$$

$$= +|\Phi_0\rangle$$

$R^m|\Phi_0\rangle = ?$

Let's find an orthonormal basis in which to analyze the action of R.

$|\vec{y}\rangle$ is the target state.

$C \equiv \langle\vec{y}|\Phi_0\rangle = \frac{1}{\sqrt{2^n}}$ (almost orthogonal)

$|\chi\rangle \equiv \frac{1}{\sqrt{\omega}}\left(|\Phi_0\rangle - c|\vec{y}\rangle\right)$ $\omega \equiv 1 - c^2$.

check orthonormality:

$$\langle\vec{y}|\chi\rangle = \frac{1}{\sqrt{\omega}}\left(\langle y|\Phi_0\rangle - c\,\langle\vec{y}|\vec{y}\rangle\right)$$

$$= \frac{1}{\sqrt{1-c^2}}\left(c - c\right) = 0$$

$$\langle\chi|\chi\rangle = \frac{1}{\sqrt{\omega}}\left(\langle\Phi_0| - c^*\langle\vec{y}|\right)\frac{1}{\sqrt{\omega}}\left(|\Phi_0\rangle - c|\vec{y}\rangle\right)$$

$$= \frac{1}{\omega}\left(1 - c\,\langle\Phi_0|\vec{y}\rangle - c^*\langle y|\Phi_0\rangle + |c|^2\langle y|y\rangle\right)$$

$$= \frac{1}{\omega}\left(1+|c|^2 - c^*c - cc^*\right)$$

$$= \frac{1}{\omega}\left(1-c^2\right) = 1$$

We move $|\Phi_0\rangle$ in the new basis:

$$|\Phi_0\rangle = \sqrt{\omega}\,|\vec{x}\rangle + c\,|\vec{y}\rangle \qquad |x\rangle = \frac{1}{\sqrt{\omega}}\left(|\Phi_0\rangle - c|\vec{y}\rangle\right)$$

$$|\psi\rangle = \cos\frac{\theta}{2}\,|\vec{x}\rangle + \sin\frac{\theta}{2}\, e^{i\varphi}\,|\vec{y}\rangle \qquad \sqrt{\omega}\,|x\rangle = |\Phi_0\rangle - c\,|\vec{y}\rangle$$

After applying R many times,

$$|\psi\rangle = |\Phi_0\rangle \Rightarrow \sin\frac{\theta_0}{2} = c$$

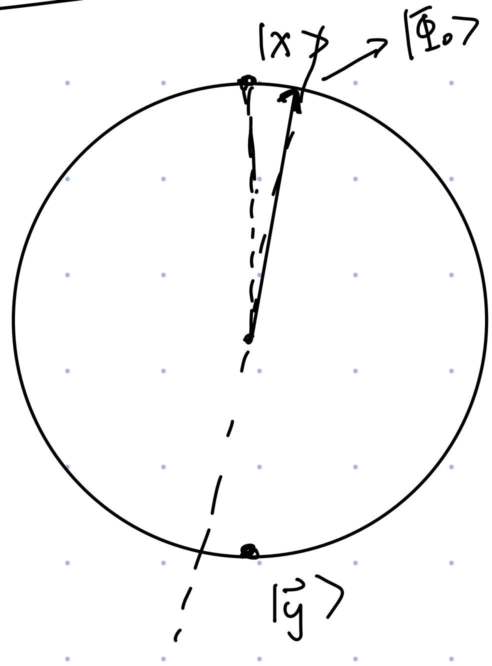
$$\rightarrow \cos\frac{\theta_0}{2} = \sqrt{\omega}$$

initial param.

$$\varphi_0 = 0.$$

Pseudo Bloch Sphere,

G is a reflection abt the Φ_0 axis.



$U_f|x\rangle = |x\rangle$, $U_f|y\rangle = -|y\rangle$ $\Rightarrow$ like Z on the pseudo Bloch sphere.

$-iZ$ = rotation by π around Z.

Z : reflection around Z.

$G U_f|x\rangle =$